

Lustjinn

Python as a backend, for a .NET developer

A course that grows with the code. This edition was built on 6 October 2026.

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CHAPTER 0

Orientation

You already know how to build a backend. You know what a request pipeline is, what dependency injection buys you, why migrations exist, how `async/await` frees a thread. None of that changes. What changes is the vocabulary, the tooling, and a handful of assumptions so deep in .NET that you no longer notice them — and those are the ones that bite.

This course follows **Lustjinn** as it is built: a web app for roleplay with a memory that does not forget, the Python successor to `airp`. Each chapter is one step of the roadmap. Claude writes the code; the chapter explains it — what each new piece is, why it is there, and what it would be in .NET. When a chapter and the code disagree, the chapter is out of date: that is a bug.

0.1 How to read this course

Every chapter has the same shape: what we are building in this step, the code that was written, the concepts it introduced, and what changed in the repository. Four kinds of box recur:

THE .NET LENS

Maps a Python idea onto the .NET one you already know — and says where the analogy breaks.

WORTH KNOWING

Background that is useful but not needed to follow the step.

WATCH OUT

A trap. Usually one that would compile — or rather, *run* — without complaint.

TRY IT

A small exercise. Optional, but the fastest way to make a concept stick.

0.2 The map

Most of the Python toolchain has a direct counterpart in what you use every day. Keep this table nearby; later chapters assume it.

You know (.NET)	Lustjinn uses (Python)	What it is
<code>dotnet CLI</code>	uv	Creates projects, installs packages, runs things
<code>nuget.org</code>	PyPI	The public package registry
<code>.csproj</code>	<code>pyproject.toml</code>	The project file: name, dependencies, tool settings
<code>packages.lock.json</code>	<code>uv.lock</code>	Exact resolved versions, committed
<code>global.json</code>	<code>.python-version</code>	Pins the runtime version
Roslyn analyzers, <code>dotnet format</code>	ruff	Linters and formatters in one
The compiler + nullable reference types	pyright (strict)	The type checker
xUnit	pytest	Tests
A fake <code>HttpMessageHandler</code>	httpx2's MockTransport	Fakes HTTP calls in tests
Kestrel	uvicorn	The web server that runs the app
ASP.NET Core minimal APIs	FastAPI	Routing, model binding, DI, OpenAPI
<code>record</code> + <code>DataAnnotations</code>	Pydantic models	Typed data with validation
<code>IOptions<T></code>	pydantic-settings	Typed configuration from the environment
EF Core	SQLAlchemy 2.0	ORM and query builder
EF Core migrations	Alembic	Schema migrations
<code>HttpClient</code>	httpx2	HTTP client, sync and async
<code>Task</code> , <code>async</code> / <code>await</code>	asyncio coroutines	Asynchronous code

The table is a starting point, not a promise. The rest of this chapter covers the four places where the mapping breaks hardest.

0.3 Break 1 — Types are checked, but not enforced

In C#, a type is a contract the compiler enforces and the runtime relies on. In Python, a type hint is an *annotation*: the interpreter stores it and otherwise ignores it.

```
def greet(name: str) -> str:
    return "Hello, " + name

greet(42)  # Runs. Fails only inside, at the "+", with a TypeError.
```

Nothing stops that call at run time. What stops it is **pyright**, run before the code ships — exactly the role the C# compiler plays, except that it is a separate tool and you could skip it. Lustjinn runs pyright in **strict** mode, in CI, on every push: that is our `TreatWarningsAsErrors`.

THE .NET LENS

`str | None` is Python's `string?`. With nullable reference types on, C# warns when you dereference something that might be null; pyright strict does the same for `None`. The difference is that C#'s `null` checks are warnings you opted into, while in strict pyright they are errors from the start.

WORTH KNOWING

Some libraries *do* read type hints at run time, on purpose. **Pydantic** and **FastAPI** are the two you will meet first: a parameter declared `limit: int` makes FastAPI parse and validate the query string as an integer and return a 422 if it is not. That is model binding, driven by the same annotations pyright checks.

0.4 Break 2 — There is no build step

`dotnet build` turns your project into an assembly; `dotnet run` runs the assembly. Python has no equivalent step you run yourself. The interpreter (**CPython**, the standard one) compiles each file to bytecode the first time it is imported, caches it in a `__pycache__` folder, and executes it. A syntax error surfaces when the file is loaded; a typo in a name surfaces only when that line runs.

That is why the tooling matters more than in .NET: **ruff** and **pyright** are the build. If they are clean and the tests pass, the code is as checked as it will get before it runs.

0.5 Break 3 — Every project gets its own interpreter environment

NuGet keeps packages in a global cache, and each project references the versions it wants; two projects can use different versions of the same package without noticing each other.

Python installs packages *into an interpreter*. Install a package into the system Python and every project on the machine sees that version. The answer is a **virtual environment**: a folder (`.venv`) holding a private copy of the interpreter's package directory, one per project.

uv manages all of it. It downloads the Python version the project pins, creates `.venv`, installs exactly what `uv.lock` says, and runs commands inside that environment:

<code>uv sync</code>	<code># make .venv match uv.lock</code>	<code>≈ dotnet restore</code>
<code>uv add httpx2</code>	<code># add a dependency, update the lock</code>	<code>≈ dotnet add package</code>
<code>uv run pytest</code>	<code># run a command inside .venv</code>	<code>≈ dotnet test</code>
<code>uv python install</code>	<code># fetch the pinned Python version</code>	<code>≈ installing an SDK</code>

WATCH OUT

Your Windows machine has an older system Python (3.9) on the `PATH`. Lustjinn never uses it: `uv` fetches the version `.python-version` asks for and keeps it separate. If a command ever behaves as if it were on 3.9, it was run with `python` instead of `uv run python`.

0.6 Break 4 — Async runs on one thread

In ASP.NET Core, each request runs on a thread-pool thread. If a handler blocks — a synchronous database call, a `Thread.Sleep` — it wastes that one thread, and the pool covers for it.

Python's `asyncio` runs every coroutine on **a single thread**, inside an **event loop**. An `await` hands control back to the loop so another request can make progress. A call that blocks without awaiting does not waste one thread; it freezes *every* request the server is handling until it returns.

```
async def handler():
    time.sleep(2)           # Blocks the event loop: the whole server stops for 2 s.
    await asyncio.sleep(2)  # Yields to the loop: other requests keep running.
```

This is why Lustjinn uses the async versions of everything that waits: **httpx2's** `AsyncClient`, SQLAlchemy's async engine, an async Postgres driver.

THE .NET LENS

`async def` $\approx$ an `async Task` method, and `await` means the same thing. The difference is what happens *between* awaits: in .NET, other work runs on other threads; in Python, other work runs only when you await. There is no `ConfigureAwait`, and no `SynchronizationContext` to think about.

WORTH KNOWING

FastAPI hides one sharp edge for you. An endpoint declared with plain `def` (not `async def`) is run on a separate thread pool, so blocking inside it is safe. An endpoint declared `async def` runs on the event loop and must never block.

0.7 Smaller differences you will meet soon

- **Files are modules; folders are packages.** There is no `namespace` keyword and no project reference: `from lustjinn.stories import Story` imports by path from the package folder.
- **Naming:** `snake_case` for functions, variables and modules; `PascalCase` for classes; `UPPER_CASE` for constants. The leading underscore (`_helper`) means "private by convention" — nothing enforces it.
- **self is explicit.** A method's first parameter is the instance, written out. The constructor is `__init__`.
- **Indentation is syntax.** Blocks have no braces; the indentation is the block. ruff's formatter keeps it consistent.
- `None`, `True`, `False` are capitalised.

0.8 What is already on your machine

The setup for this course and for step 1 is installed:

- **uv** — the Python toolchain. It will fetch the current Python on first use.
- **pandoc** and **Typst** — they turn the Markdown chapters of this course into this PDF.

The course lives in the repository under `docs/course/`. To rebuild it:

```
./docs/course/build.ps1          # the dark edition
./docs/course/build.ps1 -Light    # a light edition, for printing
```

TRY IT

Run `uv --version` and `uv python list`. The second command lists the Python versions uv can install, and any it already has. Next chapter, one of them becomes Lustjinn's.

0.9 What is next

Chapter 1 — Hello, deployed. A uv project, FastAPI with one endpoint, one pytest test, ruff and pyright in CI, and the whole thing deployed to Render — so that `GET /health` answers from the internet. It teaches the project layout, uv in practice, and how a Python web app is actually served.

CHAPTER 1

Hello, deployed

The goal of step 1 is the smallest thing that is real: a Python web app with one endpoint, one test, the quality gates, and a path to a public URL. Nothing about roleplay yet. Everything later grows out of this skeleton, so it is worth understanding every file in it.

The step landed in three pieces, and this chapter follows them in order:

1. **The project** — uv, FastAPI, one endpoint, one test, ruff and pyright.
2. **CI** — the gates running on GitHub Actions on every pull request.
3. **The deploy** — the path to Render, built and waiting: the project runs locally until it is ready for the cloud.

1.1 What uv created

One command started the project:

```
uv init --package --name lustjinn --python 3.14
```

and two more added the libraries:

```
uv add fastapi "uvicorn[standard]"
uv add --dev pytest httpx2 ruff pyright
```

The result, ignoring the docs:

<code>.python-version</code>	3.14 – which Python this project runs on
<code>pyproject.toml</code>	the project file
<code>uv.lock</code>	every package, pinned to an exact version and hash
<code>.venv/</code>	the project's private environment (gitignored)
<code>src/lustjinn/</code>	
<code>__init__.py</code>	makes the folder a package
<code>main.py</code>	the FastAPI app
<code>tests/</code>	
<code>test_health.py</code>	the one test

Python 3.14 is pinned because it is the newest stable release: 3.15 exists, but only as a release candidate. uv downloaded 3.14.7 on its own, without touching the Python 3.9 already installed on the machine.

1.1.1 Why the code lives under `src/`

The package is `src/lustjinn/`, not `lustjinn/` at the root. With the code at the root, `import lustjinn` would work in tests simply because the current folder happens to be on the import path — and a test could pass against files that would never be part of the installed app. Under `src/`, the only way to import `lustjinn` is to install it, which `uv sync` installs the project into `.venv` in *editable* mode, meaning the environment points at `src/` instead of copying it, so edits are live.

THE .NET LENS

This is the test project's `<ProjectReference>`. A test project in .NET cannot see the app's source files, only the assembly the reference builds. The `src/` layout gives Python the same discipline: tests see the package as installed, the way the server will.

1.2 The project file

`pyproject.toml` is the `.csproj`. It is TOML, a format of `[tables]` and `key = value` lines.

```
[project]
name = "lustjinn"
version = "0.1.0"
requires-python = ">=3.14"
dependencies = [
    "fastapi>=0.142.2",
    "uvicorn[standard]>=0.54.0",
]

[dependency-groups]
dev = [
    "httpx2>=2.13.1",
    "pyright>=1.1.414",
    "pytest>=9.1.1",
    "ruff>=0.16.10",
]
```

- **[project]** is the standard part every Python tool understands: name, version, the Python it needs, and the packages the app needs at run time.
- **uvicorn[standard]** — the brackets name an *extra*: optional dependencies a package offers. `standard` adds uvicorn's faster event loop and HTTP parser, and file watching for `--reload`.
- **[dependency-groups]** holds what development needs but production does not: the test runner and the gates. A production install leaves them out.
- **[build-system]** says how to turn the project into an installable package (`uv_build`).
- **[tool...]** tables configure each tool — `[tool.ruff]`, `[tool.pyright]`, `[tool.pytest...]` — all in the one file.

THE .NET LENS

`"fastapi>=0.142.2"` reads like NuGet's `Version="0.142.2"`, and both mean *at least* that version. The difference is which version gets picked: NuGet resolves the **lowest** version that satisfies every constraint, `uv` the **highest**. Either way, the lock file is what makes the build repeatable — `uv.lock` plays the role of `packages.lock.json`, and it is committed.

1.3 The app

The first version of the whole application, `src/lustjinn/main.py` (the deploy, at the end of the chapter, grows it by one field):

```
"""The FastAPI application: the object uvicorn serves."""

from fastapi import FastAPI

app = FastAPI(title="Lustjinn")

@app.get("/health")
async def health() -> dict[str, str]:
    """Answers while the process is up. Render and the PWA's waking screen poll it."""
    return {"status": "ok"}
```

Line by line:

- The string at the top of a file, function or class is a **docstring**: documentation the language keeps (`help()` shows it). It plays the part of an XML `<summary>` comment.
- `from fastapi import FastAPI` imports one name from a package — a `using` that names exactly what it brings in.
- `app = FastAPI(...)` creates the application at module level. When Python imports this file, the line runs, and `app` exists from then on.
- `@app.get("/health")` is a **decorator**. Python calls `app.get("/health")`, which returns a function; that function receives `health`, registers it as the handler for `GET /health`, and hands it back unchanged. The `@` line is shorthand for exactly that.
- `async def health() -> dict[str, str]` declares a coroutine that returns a dictionary of strings. FastAPI reads the return annotation: it documents the response in the OpenAPI schema and validates what the function returns.
- `return {"status": "ok"}` — FastAPI serialises the dictionary to JSON.

THE .NET LENS

Put side by side with a minimal API, there is almost nothing to translate:

```
var app = WebApplication.CreateBuilder(args).Build();
app.MapGet("/health", () => new Dictionary<string, string> { ["status"] = "ok" });
app.Run();
```

`app.MapGet(...)` becomes the decorator, and `Dictionary<string, string>` becomes `dict[str, str]`. What has no counterpart is `app.Run()` — see the next section.

1.4 How Python is served

A .NET app is an executable: `Program.cs` builds the app and `app.Run()` starts Kestrel inside the same process. A Python web app is not run; it is **imported by a server**. The server here is **uvicorn**:

```
uv run uvicorn lustjinn.main:app --reload
```

`lustjinn.main:app` means *import the module `lustjinn.main` and serve the object called `app` in it*. `uvicorn` owns the process, the socket and the event loop; FastAPI only answers requests handed to it. `--reload` restarts the server when a file changes — for development only.

The contract between the two is **ASGI** (Asynchronous Server Gateway Interface): a standard signature every async Python server can call and every async framework implements. That is why FastAPI could run on another ASGI server without changing a line.

THE .NET LENS

`uvicorn` is Kestrel, and ASGI is the boundary between Kestrel and the ASP.NET Core pipeline — except that in Python the boundary is a public standard, so servers and frameworks mix and match.

With the server running, FastAPI also serves its own documentation: `/docs` is an interactive Swagger UI and `/openapi.json` the schema, both generated from the code — no Swashbuckle to add.

1.5 The test

The first version of `tests/test_health.py`:

```
from fastapi.testclient import TestClient

from lustjinn.main import app

client = TestClient(app)

def test_health_answers_ok() -> None:
    response = client.get("/health")

    assert response.status_code == 200
    assert response.json() == {"status": "ok"}
```

- **pytest finds tests by name:** files called `test_*.py`, functions called `test_*`. No attribute, no test class needed.
- **Plain `assert`.** `pytest` rewrites `assert` statements when it loads a test file, so a failure shows both sides of the comparison — no `Assert.Equal` family to learn.
- **`TestClient`** runs the app in-process and speaks to it over ASGI directly: no server, no port, no network.

THE .NET LENS

`TestClient(app)` is `WebApplicationFactory<Program>` plus `CreateClient()`, in one line.

WATCH OUT

When this test was first written, the dev dependency was `httpx`, as countless tutorials say. The test passed — with a deprecation warning, because `Starlette` (the toolkit under FastAPI) now wants its successor, `httpx2`. `pyright` went further and failed: with `httpx2` missing, it could not know the types of `client.get`, `response.status_code` or `response.json`, and strict mode refuses unknown types. Two lessons in one. Tutorials age fast in Python — check what the

library itself asks for. And strict typing earns its keep on day one: it turned a quiet warning into an error.

1.6 The gates

Four commands decide whether a change is acceptable. All four are clean on this commit:

```
uv run pytest          # the tests
uv run ruff check      # lint
uv run ruff format --check # formatting, without rewriting
uv run pyright         # types, strict
```

ruff is configured in `pyproject.toml` with a deliberate set of rule families. Most are style and likely-bug checks. One deserves a mention: **ASYNC** flags blocking calls made inside `async def` functions — the event-loop freeze from chapter 0, caught before it runs.

pyright runs in `strict` mode over `src` and `tests`. Strict means every value must have a type pyright can know; an unknown type is an error, not a shrug.

TRY IT

Start the server with `uv run uvicorn lustjinn.main:app --reload` and open `http://127.0.0.1:8000/docs`. Call `/health` from the Swagger page.

Then, in `main.py`, change the return to `{"status": 1}` and run `uv run pyright`. The annotation says `dict[str, str]`; pyright says no. Change it back.

1.7 CI on GitHub Actions

The gates are only worth something if nothing reaches `develop` without passing them. A workflow in `.github/workflows/ci.yml` runs all four on GitHub, where the repository lives:

```
on:
  push:
    branches: [main]
  pull_request:
    branches: [main, develop]

concurrency:
  group: ci-${{ github.event.pull_request.number || github.run_id }}
  cancel-in-progress: ${{ github.event_name == 'pull_request' }}

jobs:
  gates:
    runs-on: ubuntu-latest
    timeout-minutes: 10
    env:
      UV_VERSION: "0.12.22"
    steps:
      - uses: actions/checkout@v5
        with:
          persist-credentials: false
      - name: Install uv
```

```

run: |
  curl -LsSf "https://astral.sh/uv/${UV_VERSION}/install.sh" | sh
  echo "$HOME/.local/bin" >> "$GITHUB_PATH"
- name: Sync the environment from uv.lock
  run: uv sync --locked
- name: ruff check
  run: uv run ruff check
- name: ruff format
  run: uv run ruff format --check
- name: pyright (strict)
  run: uv run pyright
- name: pytest
  run: uv run pytest

```

Reading it top to bottom:

- **on** — a pull request into `develop` or `main` runs the gates. A push to `develop` does **not**: `develop` only moves by merging a pull request whose gates already ran, so running them again would test the same code twice. A push to `main` — a release — does.
- **concurrency** — a new push to a pull request cancels that pull request's previous run. Only the newest commit's result matters.
- **runs-on: ubuntu-latest** — a fresh Linux virtual machine for every run. `uv` and the project's Python come from the steps below, so a green run proves the project builds from its committed files alone.
- **timeout-minutes** — a stuck job stops after ten minutes. The default is six hours.
- **Install uv** — one pinned release, the same version as on the development machine. Python then comes from `uv`, exactly as on Windows: `uv sync` reads `.python-version` and fetches 3.14.
- **uv sync --locked** — creates the environment from `uv.lock`, and **fails** if the lock no longer matches `pyproject.toml`. Without `--locked`, `uv` would quietly re-resolve, and CI would test versions nobody committed.
- **The gates, cheapest first** — a formatting slip fails in seconds, before `pyright` and the tests run.

THE .NET LENS

`uv sync --locked` is `dotnet restore --locked-mode`: restore exactly what the lock file says, or fail. And the workflow as a whole is the pipeline you would write around `dotnet build` and `dotnet test` — with `uv sync` as the restore and `pyright` and `ruff` standing in for the build.

WORTH KNOWING

The repository is private, so every minute a workflow runs is billed against the account's monthly allowance, and each job is rounded up to a whole minute. That is the reason for three of the choices above: nothing runs on a `develop` push, superseded runs are cancelled, and every job has a timeout. A run of these gates takes under a minute.

1.8 The deploy

The last piece is the path to the internet: a **Render** free web service, called `lustjinn-staging`, serving the `develop` branch. The path is built and tested as far as it can be without an account — but it has not run yet. The project stays on the development machine until it is worth putting in the cloud, and this section describes what will happen then.

1.8.1 What Render runs

Render's native Python runtime needs no Dockerfile. It reads `.python-version` to choose the interpreter, and because the repository contains `uv.lock`, it installs `uv` too. The service is described in `render.yaml` at the root of the repository — a **blueprint**:

```
services:
  - type: web
    name: lustjinn-staging
    runtime: python
    plan: free
    branch: develop
    buildCommand: uv sync --locked --no-dev
    startCommand: uv run --no-sync uvicorn lustjinn.main:app --host 0.0.0.0 --port $PORT
    healthCheckPath: /health
    autoDeployTrigger: "off"
    envVars:
      - key: UV_VERSION
        value: "0.12.22"
```

- **The build** is the same `uv sync --locked` as CI, plus `--no-dev`: `pytest`, `ruff` and `pyright` stay out of production.
- **The start** runs `uvicorn` through `uv run --no-sync` — inside the environment the build created, without checking it again. It binds to `0.0.0.0`, every network interface, because Render's proxy reaches the process from outside the container; the default, `127.0.0.1`, would accept only connections from inside. And it listens on `$PORT`, the port Render chooses.
- **healthCheckPath** — Render calls `/health` to decide whether the service is up.
- **autoDeployTrigger: "off"** — a push alone deploys nothing; a deploy is always asked for.

THE .NET LENS

`--host 0.0.0.0 --port $PORT` is Kestrel's `ASPNETCORE_URLS=http://+:$PORT` — the same lesson every container platform teaches: listen where the platform says, on every interface. And the blueprint is infrastructure as code, a very small Bicep file: the service lives in the repository, not only in a dashboard.

WATCH OUT

The free tier has two properties the whole design respects. The service **sleeps** after about fifteen minutes without requests and takes 30–60 seconds to wake — which is why the future PWA is a separate static site that opens instantly and shows “waking the server”. And it has **no persistent disk**: anything written to the filesystem disappears on the next deploy or restart. No SQLite, no files. Everything lives in the database: a local Postgres for now, Neon once the app moves to the cloud.

1.8.2 How a deploy travels

A deploy is started by hand: on GitHub, *Actions* → *CI* → *Run workflow* on `develop`, with **deploy = staging**. The same workflow that runs the gates then runs one more job, `deploy-staging`, only if they all passed:

1. **Fire the hook.** A `POST` to Render's *deploy hook* — a secret URL — starts a build of the branch.

2. **Wait for the right commit.** The job polls `/health` until the service reports *this* commit. “The service is up” is not enough: the old version is up too, until the new one replaces it.

The one-time setup — creating the service from the blueprint, the hook secret and the service URL on GitHub — is in `docs/DEPLOYMENT.md`.

1.8.3 `/health` grows a field

Step 2 of that list needs the service to say which commit it runs. Render provides it in an environment variable, `RENDER_GIT_COMMIT`, so `/health` now returns it:

```
import os
from typing import Literal

from fastapi import FastAPI
from pydantic import BaseModel

app = FastAPI(title="Lustjinn")

class Health(BaseModel):
    status: Literal["ok"]
    commit: str | None
    """The deployed commit (Render's `RENDER_GIT_COMMIT`); None when not running on Render."""

@app.get("/health")
async def health() -> Health:
    """Answers while the process is up. Render, the deploy job and the PWA poll it."""
    return Health(status="ok", commit=os.environ.get("RENDER_GIT_COMMIT"))
```

This is the first **Pydantic model**. A class that inherits `BaseModel` declares its fields as annotated class attributes; Pydantic validates them when an instance is created and serialises them to JSON. FastAPI shows the model in the OpenAPI schema under its own name, `Health`.

- `Literal["ok"]` — a type with exactly one allowed value. `Health(status="fine")` fails validation, and pyright rejects it before it runs.
- `str | None` — a string or nothing. `os.environ.get(...)` returns `None` when the variable is not set, so off Render the answer is `{"status": "ok", "commit": null}`.

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`Health` is a record `Health(string Status, string? Commit)` with validation attached. `os.environ.get` is `Environment.GetEnvironmentVariable`, which also returns null when the variable is missing. `Literal["ok"]` has no direct C# counterpart; the nearest is a one-value enum.

The tests grow with it:

```
def test_health_reports_the_deployed_commit(monkeypatch: pytest.MonkeyPatch) -> None:
    monkeypatch.setenv("RENDER_GIT_COMMIT", "3d36977")

    assert client.get("/health").json()["commit"] == "3d36977"

def test_health_reports_no_commit_off_render(monkeypatch: pytest.MonkeyPatch) -> None:
    monkeypatch.delenv("RENDER_GIT_COMMIT", raising=False)
```

```
assert client.get("/health").json() == {"status": "ok", "commit": None}
```

`monkeypatch` is a **fixture**: pytest sees a parameter with that name and passes in a ready-made object. `setenv` and `delenv` change the environment for this test only — pytest undoes both when it ends, so no test leaks state into the next.

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Fixtures are pytest's dependency injection — by parameter name, the way ASP.NET Core injects by type. xUnit's nearest equivalents are constructor injection and `IClassFixture<T>`; the automatic undo is what you would otherwise write in `Dispose()`.

TRY IT

Run the server with a commit of your own, in PowerShell:

```
$env:RENDER_GIT_COMMIT = "hello"; uv run uvicorn lustjinn.main:app
```

`/health` now reports `"commit": "hello"`, and `/docs` shows the `Health` schema. Then try `Health(status="fine", commit=None)` in `main.py` and run `uv run pyright`.

1.9 What step 1 leaves behind

- A Python 3.14 project managed by `uv`, locked, with a `src/` layout.
- A FastAPI app with one endpoint, three tests, and four gates — all clean.
- CI on GitHub Actions running the gates on every pull request.
- A deploy path to Render staging that refuses to run on red gates and checks what it deployed — ready for the day the app moves to the cloud.

Next — Chapter 2, Stories and messages. The first real data, in a Postgres running in Docker: settings from the environment, sign-in, SQLAlchemy models, the first Alembic migration, and a trigger that makes messages append-only.

CHAPTER 2

Stories and messages

Step 1 built an app that could say it was alive. Step 2 gives it something to keep: a database, the tables for stories and their messages, a rule the database enforces by itself, an API over them, and a lock on the door. Everything runs on your machine — Postgres in a Docker container, the API under uvicorn. The cloud comes later and changes one connection string.

This is the chapter where Python's backend stack stops looking like a toy and starts looking like what you already know. Almost every piece has a .NET twin; the interesting parts are the few places where the twin behaves differently.

2.1 Settings: one typed object from the environment

Before anything can connect to anything, the app needs to be told where the database is and who may sign in. `src/lustjinn/settings.py`:

```
class Settings(BaseSettings):
    model_config = SettingsConfigDict(env_prefix="LUSTJINN_", env_file=".env", extra="ignore")

    database_url: str
    username: str
    password: SecretStr
    token_secret: SecretStr = Field(min_length=32)
    token_days: int = Field(default=30, ge=1, le=365)
    cors_origins: list[str] = []

    @lru_cache
    def get_settings() -> Settings:
        return Settings()
```

`BaseSettings` is a Pydantic model that fills itself in: each field is read from the environment variable of the same name with the prefix — `LUSTJINN_DATABASE_URL`, `LUSTJINN_USERNAME` — or from a `.env` file in the working directory, real variables winning. A field with no default is required. Types are enforced: `token_days` must parse as an integer between 1 and 365.

- **SecretStr** wraps a string so that printing or logging the settings shows `*****`. You get the value out on purpose, with `.get_secret_value()`.
- **@lru_cache** on a function with no arguments turns it into “compute once, then return the same object”: the settings are read on the first call and reused after that.

- **Failing early.** The app's *lifespan* — code FastAPI runs once at start-up and once at shutdown — calls `get_settings()`. A missing password stops uvicorn before it accepts a single request, with an error naming the field.

THE .NET LENS

`Settings` is the class you would bind with `services.Configure<Settings>(config)` and receive as `IOptions<Settings>`; the environment-variable and `.env` layering is the configuration builder; `Field(min_length=32)` is a `DataAnnotation` plus `ValidateOnStart()`. The lifespan is `IHostedService.StartAsync / StopAsync` in one function, split by a `yield`.

`.env` is gitignored. `.env.example` is committed and lists every variable; copy it and fill in the blanks.

2.2 A database in a container

`docker-compose.yml` describes one service: Postgres 17 with the `pgvector` extension available, listening on port 5440 of this machine only, its data in a named volume.

```
docker compose up -d      # start it; the data survives restarts
docker compose down      # stop it; add -v to throw the data away
```

Port 5440 rather than the usual 5432 so it never collides with another project's Postgres. Postgres 17 because Neon offers it: when the app moves to the cloud, the schema and every query stay as they are.

2.3 SQLAlchemy: the engine, the session, the models

SQLAlchemy is the ORM. Three objects carry it, and each has an EF Core counterpart — though the unit of work is more explicit here.

2.3.1 The engine and the session

```
@lru_cache
def get_engine() -> AsyncEngine:
    return create_async_engine(get_settings().database_url, pool_pre_ping=True)

@lru_cache
def get_sessionmaker() -> async_sessionmaker[AsyncSession]:
    return async_sessionmaker(get_engine(), expire_on_commit=False)

async def get_session() -> AsyncGenerator[AsyncSession]:
    async with get_sessionmaker()() as session:
        yield session
```

- The **engine** owns the connection pool. One per process, created on first use.
- A **session** is one conversation with the database: it tracks the objects you loaded or added, and writes the changes when told. One per request.
- **get_session** is a generator: it yields a session to whoever asked, and the code after the `yield` — here, leaving the `async with`, which closes the session — runs when the request ends.

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The engine is the pooled connection factory behind `AddDbContext`. The session is the `DbContext`: change tracking, identity map, one unit of work, scoped to a request. Two differences will matter all through this project:

- **flush() and commit() are separate.** `flush()` sends the pending INSERTs and UPDATEs but leaves the transaction open; `commit()` ends it. EF's `SaveChanges()` does both.
- **There is no lazy loading in async code.** Touching `story.messages` cannot quietly run a query, because a query needs an `await`. The relationships are declared `lazy="raise"`, so a forgotten load fails loudly instead of deadlocking or lying. You load what you need, in the query — the discipline EF asks for with `Include`, enforced.

`expire_on_commit=False` is the other async accommodation: by default SQLAlchemy forgets an object's values at commit and reloads them on next access, which again would be a hidden query.

2.3.2 The models

`src/lustjinn/models.py`, abridged:

```
class Story(Base):
    __tablename__ = "stories"

    id: Mapped[uuid.UUID] = mapped_column(primary_key=True, default=new_id)
    name: Mapped[str] = mapped_column(String(200))
    character_id: Mapped[uuid.UUID] = mapped_column(
        ForeignKey("characters.id", ondelete="RESTRICT"), index=True
    )
    persona_id: Mapped[uuid.UUID | None] = mapped_column(
        ForeignKey("personas.id", ondelete="RESTRICT"), index=True
    )
    model: Mapped[str | None] = mapped_column(String(200))
    created_at: Mapped[datetime] = mapped_column(DateTime(timezone=True),
server_default=func.now())
    deleted_at: Mapped[datetime | None] = mapped_column(DateTime(timezone=True))
```

A model is a class whose annotated attributes are columns. `Mapped[str]` is a NOT NULL text column; `Mapped[str | None]` is nullable — the type hint is the nullability, checked by pyright and enforced by Postgres. `mapped_column(...)` adds what the type cannot say: lengths, keys, defaults, indexes.

Four tables exist after this step: `characters` and `personas` (minimal for now; the library step fills them out), `stories`, and `messages`. The decisions worth noticing:

- **Ids are UUID version 7** (`uuid.uuid7()`, new in Python 3.14): unique like any UUID, but beginning with a timestamp, so they sort by creation time and index well.
- **A story points at its character by id**, with `ON DELETE RESTRICT`. `airp` stored the character's *name*, so renaming one was impossible, and "not while a story uses it" was checked in code. Here the database refuses the delete itself.
- **Deleting is hiding**. `deleted_at` is set and every query filters on it. The row stays.
- **A message has a sequence**, unique within its story, and a `request_hash` that is unique within its story *among rows that have one* — a **partial index**. Step 3 uses the hash to make a retried request harmless.
- **model is null when a person wrote the message**. The reader's turns, and a story's opening. A later step reads that null to know an opening must not be rerolled.

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`Mapped[...]` and `mapped_column` are the entity class and its `OnModelCreating` configuration merged into one declaration. `ForeignKey(..., onDelete="RESTRICT")` is `.onDelete(DeleteBehavior.Restrict)`. The partial index is `HasIndex(...).HasFilter("...")`.

2.4 Alembic: the migrations

The models say what the schema should be. **Alembic** gets a database there, one versioned step at a time, and records in a table (`alembic_version`) how far it has gone.

```
uv run alembic revision --autogenerate -m "what changes" # write a migration from the models
uv run alembic upgrade head                             # apply what is not applied yet
uv run alembic downgrade -1                             # undo the last one
```

Autogenerate compares the models with a live database and writes the difference as Python: `op.create_table(...)`, `op.create_index(...)`. The result is a draft — it is read, tidied and committed like any code. `migrations/versions/0001_stories_and_messages.py` began that way. `migrations/env.py` is the file Alembic runs for every command. Ours does two things: it takes the database URL from `get_settings()`, so there is one source of truth, and it accepts a connection handed in from outside — which is how the tests run the real migrations against their own database.

THE .NET LENS

`alembic revision --autogenerate` is `dotnet ef migrations add`; `alembic upgrade head` is `dotnet ef database update`; `alembic_version` is `__EFMigrationsHistory`. The difference: there is no model snapshot file. Alembic compares against a real database every time, so the database must be at the previous migration when you generate the next.

One test keeps the two descriptions honest: it applies every migration to an empty database and asks Alembic whether anything still differs from the models. A model edited without a migration fails the build.

2.5 A rule the database enforces: messages are append-only

A story's memory is derived from its transcript — summaries, facts, embeddings. If a message could be edited or deleted, everything built from it would silently become a lie. `airp` guarded this in `SaveChanges`. That protects against the application; it does nothing about a script, a console session, or a second client. So here the rule lives where the data lives.

Migration `0002` is hand-written SQL, because a trigger is not something a model can express:

```
CREATE FUNCTION messages_append_only() RETURNS trigger
LANGUAGE plpgsql AS $$
BEGIN
    IF TG_OP = 'DELETE' THEN
        IF current_setting('lustjinn.purging', true) = 'on' THEN
            RETURN OLD;
        END IF;
        RAISE EXCEPTION 'messages are append-only: a message is hidden with deleted_at, ...'
```

```

        USING ERRCODE = 'restrict_violation';
    END IF;

    IF NEW.text IS DISTINCT FROM OLD.text
    OR NEW.role IS DISTINCT FROM OLD.role
    OR NEW.story_id IS DISTINCT FROM OLD.story_id
    OR NEW.sequence IS DISTINCT FROM OLD.sequence THEN
        RAISE EXCEPTION 'messages are append-only: text, role, story and sequence cannot ...'
        USING ERRCODE = 'restrict_violation';
    END IF;
    RETURN NEW;
END
$$;

CREATE TRIGGER messages_append_only
BEFORE UPDATE OR DELETE ON messages
FOR EACH ROW EXECUTE FUNCTION messages_append_only();

```

A trigger function runs for each affected row, before the change. `OLD` and `NEW` are the row before and after; `RAISE EXCEPTION` aborts the statement. Reading it:

- **A delete is refused** — unless the transaction has announced a purge with `SET LOCAL lustjinn.purging = 'on'`. That is the one door, for the day a story is erased for good, and it closes by itself when the transaction ends.
- **What a turn said, who said it and where it sits cannot change** — purge or no purge.
- **Everything else may**: `deleted_at`, to hide a message; `request_hash`; columns later steps add.
- A second trigger refuses `TRUNCATE`, which empties a table without visiting rows and so never fires the first.

The error code is one of Postgres's own integrity codes, so SQLAlchemy raises `IntegrityError` — the same exception as for a broken foreign key. To the application the trigger is one more constraint.

WORTH KNOWING

`IS DISTINCT FROM` is `<>` that treats `NULL` as a value: `NULL <> NULL` is unknown, while `NULL IS DISTINCT FROM NULL` is false. In a guard you want the second.

2.6 Dependency injection, FastAPI's way

Every endpoint that touches the database needs a session. It asks for one in its signature:

```

Session = Annotated[AsyncSession, Depends(get_session)]

@router.get("")
async def list_stories(session: Session) -> List[StoryOut]:
    rows = await session.execute(_stories())
    return [_out(*row) for row in rows]

```

`Depends(get_session)` tells FastAPI: before calling this function, call `get_session` and pass in what it yields. `Annotated[X, ...]` attaches that instruction to the type, and the alias `Session` gives the pair a name so every endpoint can write `session: Session`.

There is no container and no registration step. A dependency is any callable; its own parameters can be dependencies in turn; a generator dependency gets its cleanup run after the response.

THE .NET LENS

`Depends(get_session)` is constructor injection of a scoped service, declared at the point of use instead of in `Program.cs`. The lifetime is per request, like `AddScoped`. What replaces `ConfigureTestServices` is `app.dependency_overrides[get_session] = something_else` — a dictionary from the real provider to its stand-in. The tests use exactly that.

2.7 The stories API

`src/lustjinn/stories.py` holds five endpoints under `/stories`: create, list, read one with its messages, rename, delete. Three things in it are new.

Pydantic models are the DTOs. What comes in and what goes out are declared as classes, and FastAPI validates the first and serialises the second:

```
Name = Annotated[str, StringConstraints(strip_whitespace=True, min_length=1, max_length=200)]

class NewStory(BaseModel):
    name: Name
    character_id: uuid.UUID
    persona_id: uuid.UUID | None = None
    model: ModelName | None = None
```

A body with a blank name, or an id that is not a UUID, never reaches the function: FastAPI answers 422 with the field and the reason.

A router groups endpoints. `APIRouter(prefix="/stories")` is `MapGroup("/stories")`; the app includes it with `app.include_router(...)`.

The list is one query. For each story it needs the newest visible message — its time, to sort by, and its first 200 characters, to show under the name. That is a **LATERAL** subquery: one that runs per outer row and may refer to it. EF Core produces the same SQL from a correlated `OrderByDescending(...).FirstOrDefault()`; here it is written out.

Creating a story does one more thing: if its character has an **opening**, the opening becomes message 1, role `assistant`, with no model — a page a person wrote, not a reply to regenerate.

2.8 Sign-in

The API will one day sit on a public URL with one user's private stories behind it.

`src/lustjinn/auth.py`:

1. `POST /auth/sign-in` takes the username and password and compares them with the settings.
2. If they match, it returns a **token**: a small signed document saying who it is for and when it expires (30 days by default).
3. Every other request carries `Authorization: Bearer <token>`. A dependency, `require_user`, checks the signature and the expiry before the endpoint runs.

The token is a **JWT** signed with HMAC-SHA256 using `LUSTJINN_TOKEN_SECRET`. Nothing is stored on the server: the signature proves the token was issued here, and the expiry is inside it. Changing the secret invalidates every token at once — that is “sign out everywhere”.

The guard is attached where the router is included, not on each endpoint:

```
app.include_router(auth.router)
# Everything below needs a valid token.
app.include_router(stories.router, dependencies=[Depends(auth.require_user)])
```

THE .NET LENS

This is `AddAuthentication().AddJwtBearer()` plus `MapGroup("/stories").RequireAuthorization()`, without the middleware: the check is an ordinary dependency. `HTTPBearer` is the piece that reads the header and puts an *Authorize* button on the `/docs` page.

Details that are easy to get wrong, each pinned by a test:

- **Comparison in constant time** (`secrets.compare_digest`), and both name and password are checked before answering, so neither the reply nor its timing says which one was wrong.
- **The accepted algorithm is named explicitly**. A forged token whose header says “no signature needed” is refused because the server never asks the token how to verify it.
- **A token without an expiry is refused**, as is one issued for a different username.
- **One test covers every route**: it reads the app’s own OpenAPI description and requires a 401 from each path except `/health` and `/auth/sign-in`. An endpoint added next month is guarded, or that test fails.

WATCH OUT

FastAPI’s default 422 response includes the input it rejected. For a malformed sign-in that input is the password; for a story it is the text. A test caught the first, and the app now answers validation errors with the field and the problem only. Defaults are written for tutorials — read what yours actually send.

Two more things ride along. **CORS** — which browser origins may call the API — comes from the settings and allows none by default; the web client’s origin gets added when it exists. And every response carries four headers saying it is private: not to be cached, indexed, or referred from.

2.9 Testing against a real database

The trigger, the constraints and the queries are Postgres behaviour. Faking the database would test the fake. So the tests use the real thing — and never the data you are playing with.

`tests/conftest.py` builds it from **fixtures**, each asking for the one before by naming it as a parameter:

1. **database_url** (once per run) creates an empty database with a random name on the same server, and drops it at the end.
2. **engine** (once per run) applies every Alembic migration to it — the real migrations, so they are tested on every run.

3. **session** (per test) opens a transaction that is **never committed**, and rolls it back when the test ends. The code under test may call `commit()` freely: the session turns those into savepoints inside the outer transaction.
4. **client** (per test) is an HTTP client wired straight to the app in-process, with the app's `get_session` and `get_settings` overridden to hand out the test's own.

```
async def test_deleting_a_story_hides_it_and_keeps_its_rows(
    client: httpx2.AsyncClient, session: AsyncSession
) -> None:
    story = await a_story(session)
    await session.commit()

    response = await client.delete(f"/stories/{story.id}")

    assert response.status_code == 204
    assert (await client.get(f"/stories/{story.id}")).status_code == 404
    kept = await session.get(Story, story.id, populate_existing=True)
    assert kept is not None and kept.deleted_at is not None
```

The test arranges through the session, acts through HTTP, and checks both the response and the row — all inside one transaction that leaves nothing behind.

THE .NET LENS

`scope="session"` fixtures are `IClassFixture` /collection fixtures; the per-test rollback is what `Respawn` or a transaction scope gives you in .NET; the in-process client is `WebApplicationFactory.CreateClient()`. `async def` tests need a plugin, `pytest-asyncio`, configured here to run everything on one event loop — the engine's connections belong to it.

Tests that need a character use the **dummy**: "The Gilded Heron", a fictional card of about 1,850 words with an opening and a persona, under `tests/fixtures/dummy/`. `airp`'s tests used a four-word card inline and missed a bug that cost a real story twenty-four turns of memory; from step 5 on, the memory tests here run on a card the size of a real one.

In CI the same Postgres image runs as a *service container* beside the job, on the same port, so the tests' default connection string works there unchanged.

TRY IT

Bring it all up and use it:

```
cp .env.example .env # fill in a username, a password, a token secret
docker compose up -d
uv run alembic upgrade head
uv run python scripts/seed_dummy.py # prints the dummy character's id
uv run uvicorn lustjinn.main:app --reload
```

Open `http://127.0.0.1:8000/docs`. Call `POST /auth/sign-in`, copy the `access_token`, press **Authorize** and paste it. Then `POST /stories` with a name and the character id the seed script printed — the response's first message is the opening.

Now try to break the rule. In another terminal:

```
docker compose exec db psql -U lustjinn -c "UPDATE messages SET text = 'rewritten'"
docker compose exec db psql -U lustjinn -c "DELETE FROM messages"
```

Both are refused, by the database, with no application running in between.

2.10 What step 2 leaves behind

- Typed settings that stop the app at start-up when something is missing.
- A local Postgres, four tables, two migrations, and a test that keeps models and migrations in step.
- Messages that cannot be rewritten or deleted, whoever asks.
- A stories API behind a bearer token, with every route's guard tested from the app's own schema.
- A test suite that runs against real Postgres and leaves no trace.

Next — Chapter 3, One turn against OpenRouter. The first conversation with a model: an async HTTP client, the reply streamed to the caller as it is written, the reader's turn saved before the model is asked, and a ledger row for what it cost.

CHAPTER 3

One turn against OpenRouter

Step 2 left a story that could be opened and could not be played. Step 3 plays it: the reader sends a message, the API asks a model for the reply and streams it back word by word, and both sides of the exchange are kept — the reader's turn *before* the model is asked, the reply with the model that wrote it, the host that served it, and what it cost. Around that core, three pieces of turn mechanics: a retry that cannot store a message twice, a reroll that hides the old reply and keeps it, and slash commands, where a typo is refused instead of billed.

This chapter is about **async I/O**: an HTTP client that streams, a response that streams, and the generators that connect the two. It is also where money enters the code, and where a provider's failure modes get names.

3.1 One endpoint, one key

Every model call goes to **OpenRouter**, which fronts many models behind one OpenAI-compatible endpoint and one key. Choosing a model is configuration; so is pointing at a different provider that speaks the same shape. The settings grow accordingly (`src/lustjinn/settings.py`):

```
openrouter_api_key: SecretStr | None = None
openrouter_base_url: str = "https://openrouter.ai/api/v1"
model: str = "deepseek/deepseek-v4-flash"
temperature: float = Field(default=1.0, ge=0, le=2)
max_tokens: int = Field(default=1024, ge=1, le=32768)
model_timeout_seconds: int = Field(default=180, ge=1, le=3600)
prefer_providers: list[str] = []
ignore_providers: list[str] = []
allow_provider_fallbacks: bool | None = None
think_before_replying: bool = False
```

The key is **optional** here on purpose. The app must start without one — the stories API has nothing to do with the model — and a model call without a key fails before anything is sent, with an error naming the variable. `temperature` and `max_tokens` carry ranges, like `token_days` did.

`think_before_replying` is off, which sends OpenRouter's `reasoning: {enabled: false}` on every reply: without it, the default model came back empty three times in five, every token spent thinking.

The two provider lists are the one setting that makes a measurable difference to the bill. OpenRouter spreads a model across hosts, and they are not interchangeable: some cache prompts, some do not, some return garbage. The lists go out as OpenRouter's `provider` field, trimmed and otherwise exactly as typed — the router drops a slug it does not know without complaint, so lower-casing to be helpful would only hide the typo.

3.2 Streaming, in both directions

A reply takes seconds to write. Showing it as it is written is the difference between an app and a form, so the model is asked to stream — and the API streams to its own clients in turn. Both use the same format, **server-sent events**: one long HTTP response of `text/event-stream`, carrying events a few lines at a time, each ended by a blank line.

```
event: delta
data: {"text": "She looks up"}

: a line starting with a colon is a comment – OpenRouter sends these as keep-alives

data: [DONE]
```

`src/lustjinn/sse.py` reads it and writes it, in about forty lines:

```
async def events(lines: AsyncIterable[str]) -> AsyncIterator[Event]:
    name: str | None = None
    data: list[str] = []
    async for raw in lines:
        line = raw.rstrip("\r\n")
        if not line:
            if data:
                yield Event("\n".join(data), name)
                name, data = None, []
            elif line.startswith(":"):
                continue
            else:
                field, _, value = line.partition(":")
                value = value.removeprefix(" ")
                if field == "data":
                    data.append(value)
                elif field == "event":
                    name = value
        if data:
            yield Event("\n".join(data), name)
```

This is the first **async generator**: an `async def` with `yield` in it. Calling it runs nothing; it returns an object that produces one `Event` each time the caller asks with `async for`, and between two events the function is suspended exactly where its `yield` is, holding its local variables. The `lines` it reads are themselves produced the same way, by the HTTP response, as bytes arrive on the socket.

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An async generator is a method returning `IAsyncEnumerable<T>` with `yield return` inside, and `async for` is `await foreach`. The mental model carries over whole: lazy, pull-based, suspended between items. One difference — there is no `IAsyncDisposable` on the consumer's side to think about; leaving the `async for` early is enough, and `aclose()` on the generator is called for you when it is garbage-collected or the `async with` it lives in ends.

WATCH OUT

The orientation chapter's map said **respx** fakes HTTP in tests. It does not any more: **respx** supports `httpx`, and this project moved to **httpx2** in step 1, when Starlette asked for it. The map

is corrected in this edition. The tests fake the network with `httpx2`'s own `MockTransport` instead, which turns out to be simpler — a function from request to response, the exact shape of a fake `HttpMessageHandler`.

3.3 The client

`src/lustjinn/openrouter.py` holds one class with three methods: `stream`, which does the work; `complete`, which gathers a stream into one reply for callers with nothing to show while it is written; and `models`, which lists what OpenRouter serves. The heart of it:

```
async def stream(self, messages, *, model=None, temperature=None, max_tokens=None,
                  frequency_penalty=None, reasoning=None) -> AsyncIterator[str | Reply]:
    body: dict[str, object] = {
        "model": model or settings.model,
        "stream": True,
        "temperature": ...,
        "max_tokens": ...,
        "messages": [{"role": m.role, "content": m.content} for m in messages],
    }
    if frequency_penalty is not None:
        body["frequency_penalty"] = frequency_penalty
    if reasoning is not None:
        body["reasoning"] = {"enabled": reasoning}
    if (routing := self._routing()) is not None:
        body["provider"] = routing

    text: list[str] = []
    async with self._request("POST", "chat/completions", body) as response:
        async for event in sse.events(response.aiter_lines()):
            if event.data == "[DONE]":
                break
            chunk = _parse(event.data)
            ...
            for choice in _objects(chunk, "choices"):
                delta = _object(choice, "delta") or {}
                if piece := _string(delta, "content"):
                    text.append(piece)
                    yield piece
            usage = _object(chunk, "usage") or usage

    yield Reply(text="".join(text), model=..., provider=..., cost=_decimal(usage, "cost"), ...)
```

Things to notice, in the order they appear.

Keyword-only parameters. The `*` in the signature means everything after it must be passed by name: `stream(messages, temperature=0.4)`, never `stream(messages, 0.4)`. The donor project had three bugs from a positional argument added in the middle of a long list; here that mistake cannot be typed.

Fields sent only when asked for. `frequency_penalty` absent and `frequency_penalty: 0` are different requests to some backends; only absence means “no opinion”. `reasoning` and `provider` are OpenRouter’s own fields, so a backend that refuses unknown keys never sees them unless something is configured. The walrus `:=` assigns and tests in one expression, the way `if (Routing() is { } routing)` does in C#.

The stream yields two kinds of thing. Pieces of text as they arrive, then — once — a `Reply` with the whole text and the accounting. The return type says so: `AsyncIterator[str | Reply]`. The caller tells them apart with `isinstance`, which pyright uses to narrow the type on each branch.

```
async for piece in openrouter.stream(messages, ...):
    if isinstance(piece, Reply):
        written = piece
    else:
        yield format_event("delta", {"text": piece})
```

THE .NET LENS

`str | Reply` is a union type: a value that is one or the other, with the checker tracking which. C# has no unions; the nearest is a `OneOf<string, Reply>` or a base record with two subtypes and a type pattern in a `switch`. `isinstance` narrowing is the `is` pattern: after `if (piece is Reply written)`, `written` is a `Reply` — pyright does the same without the second name.

Reply is a frozen dataclass. `@dataclass(frozen=True, kw_only=True)` generates the constructor, equality and `repr` from the annotated fields, refuses assignment after construction, and requires every field by name. It is a `record` with `init`-only properties. Its `cost` field is a `Decimal`.

3.3.1 Money is a `Decimal`

JSON has one number type, and Python reads it as `float` by default: `0.0002` arrives as the nearest binary fraction, and a few hundred of those summed give `0.0006000000000000001`. A report about money that has to apologise for its own arithmetic is not one anybody checks against an invoice. So the parser reads every number with a decimal point as `Decimal`:

```
parsed: object = json.loads(text, parse_float=Decimal)
```

and the cost is converted nowhere else. In Postgres it is a `NUMERIC(18, 10)`, and SQLAlchemy maps that to `Decimal` on the way back.

THE .NET LENS

`decimal.Decimal` is `System.Decimal`, with one difference that matters here: it is arbitrary precision rather than 96 bits, and its precision is a context setting rather than fixed. Nothing in this project is affected; everything is ten decimal places or fewer.

3.3.2 Reading `Any` without lying to the checker

`json.loads` returns `Any` — a value the checker gives up on. In strict mode every use of it is an error, which is the point: a key that is not there or a value of the wrong type should be a decision, not a crash in a log. Four small functions make the decisions, once:

```
Json = dict[str, object]

def _object(source: Json, key: str) -> Json | None:
    value = source.get(key)
    return value if isinstance(value, dict) else None
```

```
def _string(source: Json, key: str) -> str | None:
    value = source.get(key)
    return value if isinstance(value, str) else None
```

Everything the client reads goes through them, so a response shaped differently than expected produces `None` fields, never an exception — and the one place that must not be `None`, the text, is checked explicitly.

THE .NET LENS

`JsonNode` with `?["key"]?.GetValue<string>()` does the same job in the donor, null-conditional all the way down. The Python version is more explicit because `Any` is the one type the checker refuses to reason about: it has to be narrowed by hand, with `isinstance`, into something it can.

3.3.3 Every failure has one shape

The model can fail in six ways, and the client turns each into one `ModelError` whose message is fit to show the reader:

What happened	What the error says
No key configured	Names the variable. Fails before any request
Timeout, connection refused	"did not answer within 180 s", "could not reach ..."
A 4xx or 5xx	The status, and <code>error.message</code> from the body — never the body, which echoes the prompt
A 200 that is not a stream	"not a stream (text/html)": a gateway's error page
An error chunk mid-stream	What the chunk said. A host dying mid-reply looks like this
A 200 with no text	<code>finish_reason</code> , the host, and "reasoning only" when the model spent everything thinking

The last one is the subtle one. A reply that is an empty string would be stored as though the model had answered with silence — a turn that never happened, permanent and billed. So no text is a failure, and the message says which of three different problems it was: a content filter (`finish_reason: content_filter`), a ceiling hit before the first token (`length`), or a host with nothing to say (`stop`).

The error also carries `no_such_model`, true only for a 404 or a 400 saying "not a valid model". Step 7 falls back to the default model on exactly that and on nothing else — a rejected key or an empty account would refuse the default identically.

One timeout, in one place. The donor had two — its own and `HttpClient`'s 100-second default — and the wrong one fired first. Here `httpx2.Timeout(settings.model_timeout_seconds)` is set on the request and nowhere else. For a stream it bounds the connection and the gap between one chunk and the next: a reply that keeps arriving is alive however long it takes as a whole.

3.3.4 An `async with` of our own

The request is opened in a class with `__aenter__` and `__aexit__`: `enter` sends the request, checks the status and hands back the open response; `exit` closes it, whether the stream was read to the end or abandoned at an exception. `async with self._request(...) as response:` is the whole of the caller's responsibility.

THE .NET LENS

`__aenter__` / `__aexit__` are `IAsyncDisposable` plus a factory, used with `await using`. Python splits "acquire" and "release" into two methods on the same object, and the `with` statement guarantees the second runs — the same `try/finally` a `using` expands to.

3.4 The ledger

Four kinds of call spend money: a reply, a question asked out of character, and (from step 6) a summary and a fact extraction. Two of those fire without the reader asking for anything, and were invisible in the donor for as long as spending was inferred from message rows. So every billed call writes one row to `spend`, in `src/lustjinn/ledger.py`, whatever becomes of what it produced:

```
def row(story_id, kind: SpendKind, reply: Reply, message_id=None) -> Spend:
    return Spend(story_id=story_id, kind=kind, message_id=message_id,
                  model=reply.model, provider=reply.provider, generation_id=reply.generation_id,
                  prompt_tokens=reply.prompt_tokens, completion_tokens=reply.completion_tokens,
                  cached_tokens=reply.cached_tokens, cache_write_tokens=reply.cache_write_tokens,
                  cost=reply.cost)
```

Three decisions are in the table itself:

- **cost is nullable.** Null is "the API did not say what it charged"; zero is a price. A report that added them together would be confidently wrong instead of honestly incomplete.
- **No foreign keys.** Not to the story, not to the message. Erasing a story for good (step 7) takes its messages and keeps its ledger: what the router charged exists nowhere else once the response is gone.
- **Insert-only**, by a trigger like the one on messages — but with no purge flag. Migration 0003 refuses `UPDATE`, `DELETE` and `TRUNCATE` outright.

A reply's row points at its message, which is how a later cost report will know the reply was rerolled away: by reading the message's `deleted_at` at report time, never by rewriting the row.

WORTH KNOWING

The ledger is the one table that is not derived from anything. Everything else — summaries, facts, embeddings — could be rebuilt from the transcript. The bill could not.

3.5 The turn

`POST /stories/{id}/send` with `{"text": "I come in from the rain."}`. What comes back is the stream: `delta` events while the reply is written, then one last event that says how it ended.

```

event: delta
data: {"text": "She looks up from the ledger"}

event: delta
data: {"text": ", and does not smile."}

event: done
data: {"kind": "turn", "sent": {...}, "reply": {...}, "replayed": false}

```

The last event is `done` with both messages as stored, or `error` with a reason and — this is the important part — the reader’s message, which was kept. Once a response has started streaming its status code cannot change, so every failure after that point is an event in the stream, not an HTTP status. A 4xx before the stream (a blank message, an unknown command, a story that does not exist) is still a plain JSON error.

3.5.1 Persist first

The order inside `turn()` in `src/lustjinn/turns.py` is the lesson the donor paid for:

1. Compute the request’s hash and look it up (next section).
2. If the message is new, **insert it and commit** — before the model is called.
3. Build the prompt, call the model, stream the deltas.
4. On a reply: insert the assistant’s message and its ledger row, **one commit**.
5. On a failure: nothing to undo. The event says *“Your message was kept ... Do not send it again.”*

If the send were only stored after a successful reply, a provider outage would silently swallow what the reader typed — and telling them it failed outright is how the same message gets typed a second time. A test pins each half: `test_the_message_survives_a_model_that_fails` and `test_retrying_a_send_the_model_failed_does_not_store_it_twice`.

3.5.2 Idempotency, anchored on the last reply

A retry must find the message it is a retry of. The message carries a **request hash**:

```

def request_hash(story_id, anchor, text, instruction=None) -> str:
    directed = text if not instruction else f"{text}\x1f{instruction.strip()}"
    digest = hashlib.sha256(f"{story_id}|{anchor}|{directed}".encode()).hexdigest()
    return digest[:32].upper()

```

The **anchor** is the sequence number of the last visible reply — the state the message answers — and not the next free position. After a send fails at the model the reader’s turn is already stored, so the next free position has moved; anchoring on it would give the retry a different hash and store the sentence twice. Anchored on the reply, the retry hashes the same, finds the unanswered row, and asks again against it. And the same words typed again *after* a reply has landed anchor differently: saying “Hello.” twice in one story is ordinary, not a retry.

Four cases, in the order the code checks them:

The lookup finds...	What happens
Nothing	Insert the message, commit, call the model
A hidden message	Clear its hash, treat as nothing: re-sending words the reader deleted is how a turn is taken back and asked for again

The lookup finds...	What happens
A message with a visible reply after it	Return the stored pair, <code>replayed: true</code> , no model call
A message with no reply	Reuse the row and call again

The third case is reachable only by two copies of one request in flight, since the anchor moves once a reply lands; the test pins the hash with `monkeypatch.setattr` to simulate it. The second case came from a real story: the reader deleted their last turn and pasted it again, the hash matched the tombstone, and the model kept answering the story without that turn.

THE .NET LENS

`hashlib.sha256(...).hexdigest()` is `Convert.ToHexString(SHA256.HashData(...))`. The `\x1f` separator between text and instruction is a control character because a pipe or a space can appear in prose, and a separator the message can contain lets two different sends hash alike.

3.5.3 The response is a generator too

The endpoint returns a `StreamingResponse` wrapping an async generator; FastAPI iterates it and writes each string to the socket as it is produced. Two details:

```
Session = Annotated[AsyncSession, Depends(get_session, scope="request")]
```

`scope="request"` keeps the session dependency open until the response has been sent. The default scope closes a `yield` dependency when the endpoint *function* returns — which, for a streaming response, is before the stream has run. The generator would find its session closed. This is one of the few places where FastAPI's model of a request differs from the pipeline you know, and it is worth remembering whenever a response outlives its handler.

The other is a header, `X-Accel-Buffering: no`, which tells a proxy in front (Render's, nginx) to pass each event on as it comes instead of collecting the body first.

3.5.4 The prompt, for now

Step 5 builds the context builder: layers in order, a token budget, the audit. Until then `src/lustjinn/prompt.py` sends the first two layers and the last, in the order it will keep: the character's card as a system message, the persona behind its frame ("The user is playing the following person..."), the visible transcript as `user` and `assistant` turns, and, when there is one, an instruction for this call only. The instruction goes as `user` when the transcript ends on a reply and as `system` when it ends on the reader's own turn — two user turns in a row, and a model tends to answer the second and forget the first.

A story that names its own model plays on it, with the temperature mapped onto that model's measured range: the roleplay finetunes write cleanly up to 0.9 and turn to token soup at 1.3, so 1.0 on the default's scale lands at 0.6 on theirs. `temperature_for` in the client does the arithmetic; it is pure, and tested as such.

3.6 Reroll

POST `/stories/{id}/reroll` writes the newest reply again. The old one is **hidden before the call** — a model shown its last attempt writes it again — and if nothing arrives to replace it, shown again, so a failed reroll never leaves the story shorter than it found it. Hidden, not removed: the row stays, and so does its ledger row. Both calls were charged.

Two refusals, both 409: there is no reply to reroll (the newest visible message is the reader's, or there is none), and the reply is the **opening**. The opening is a page a person wrote, and it sits at the top of the story for the rest of its life; a reroll there would trade it for a guess. It is told apart by two things, because either alone is wrong: no model wrote it (`model` is null — the reason step 2 stored that null), and the reader has not taken a turn yet. A beat the model writes after the opening with no reader turn (carry on, step 7) satisfies the first and not the second, and is rerolled.

3.7 Slash commands, and `/ask`

A message is permanent and billed. (OOC: `skip to the evening`) typed into one reaches the model, is stored forever, is counted in every later prompt, and may be summarised as something that happened. A command routes the same words where they belong — or nowhere — and leaves the transcript alone. The rule that follows: **an unrecognised command is refused, never sent**. A typo would otherwise cost what the message would have cost and land in the story as nonsense the character has to react to, and append-only means it could not be taken back.

`src/lustjinn/commands.py` reads what was typed and says what it is:

```
@dataclass(frozen=True)
class Prose:
    text: str

@dataclass(frozen=True)
class Command:
    spec: Spec
    argument: str

@dataclass(frozen=True)
class Unknown:
    name: str

@dataclass(frozen=True)
class Incomplete:
    spec: Spec

Parsed = Prose | Command | Unknown | Incomplete
```

Four small records and a union of them: the parse result is one of four things, and the caller has to say what it does with each. The endpoint does so with a `match` statement:

```
match commands.parse(body.text):
    case commands.Prose(text):
        return _streamed(turn(session, openrouter, settings, story, text))
    case commands.Command(spec=commands.Spec(name="ask"), argument=question):
        return _streamed(ask(session, openrouter, settings, story, question))
    case commands.Incomplete(spec=spec):
        raise HTTPException(422, f"Usage: {spec.usage} - nothing was stored.")
```

```
case commands.Unknown(name=name):
    raise HTTPException(422, f"/{name} is not a command, so nothing was sent ...")
```

THE .NET LENS

This is a `switch` expression with type patterns and property patterns:

`case Command { Spec.Name: "ask" } c`. Python's version matches on the class, then on positional fields (`Prose(text)` binds the first field) or named ones (`spec=...`). Dataclasses get the positional form for free. The union `Parsed` is the closed set of cases a discriminated union would give you, minus the compiler's exhaustiveness check — pyright does not insist every case is handled, so the catch-all is the author's job.

The parser itself follows rules the donor arrived at one bug at a time: only a slash in the very first column counts (a slash mid-sentence is punctuation — `and/or`, a date — and prose is far more common than commands); `//` sends a line that genuinely begins with one, with the first slash stripped; the name runs to the first whitespace, a newline included, so a command whose argument starts on the next line is still that command; a command that needs an argument and has none is refused quoting its usage. The table of commands holds one entry for now, `ask`; each later step adds its own as it builds them, and until then `/do` is unknown here, and refused.

3.7.1 A question is not a turn

`/ask Where does she live?` asks the model a question about the story, out of character. The rules, each pinned by a test in `tests/test_asides.py`:

- **The same prompt as a turn**, byte for byte, up to the instruction — then the ask directive, which says what it is ("Step out of the scene. This is a question from the reader... it is not a turn"), what the answer must be (brief, plain, in the author's voice), and that it must come only from what was given: "Where it does not say, say that it does not say." Every instruction this project sends says what it is; a bare directive gets echoed back as the reply.
- **Cold and short**: temperature 0.4, 600 tokens, reasoning off, no frequency penalty. An answer can be promoted to a pinned fact in step 7, so an embellishment would become something the character believes.
- **Stored apart**. The answer streams back like a reply and goes into a new table, `asides` (migration 0004), with the model, the host, the token counts and the newest message's sequence at the time. Nothing goes into `messages`, so no later prompt — and later, no summary, no fact, no embedding — ever sees it. A ledger row of kind `aside` is written, because a billed call that left no trace would make the audit quietly stop adding up.

On a caching host the question is nearly free: the prefix is what the next turn is about to send anyway.

3.8 Testing a stream

The model is faked at the network, not above it. `tests/scripted_model.py` is a queue of answers in OpenRouter's own stream format — a keep-alive comment, the text split mid-sentence across two chunks (a client that only handled whole words would pass with one chunk and garble a real reply), a chunk with `finish_reason`, a last chunk with `usage` and `cost`, then `[DONE]` — behind an `httpx2.MockTransport`:

```
def transport(self) -> httpx2.MockTransport:
    def handle(request: httpx2.Request) -> httpx2.Response:
        self.requests.append(request)
        self.calls.append(json.loads(request.content))
        return self._answers.popleft()
    return httpx2.MockTransport(handle)
```

A test says what the model answers before the call that needs it, then asserts on what was sent:

```
model.fail().says("I could say the same about you.")

first = await send(client, story.id, "Hello.")
second = await send(client, story.id, "Hello.")

assert first.last[0] == "error"
assert second.done["reply"]["text"] == "I could say the same about you."
assert len(model.calls) == 2
```

`model` is a fixture, and the `client` fixture overrides `get_openrouter` with a client built on the scripted transport — the same `dependency_overrides` trick as the session. The vocabulary (`says`, `says_unpriced`, `fails`, `empty`, `truncated`, `has_no_such_model`) is the donor's, because its tests are the specification: every idempotency case in `tests/test_turns.py` has a namesake in `custom-airp`. `tests/streaming.py` reads the API's own stream the way a client will — through the same `sse.events` parser the client uses to read OpenRouter — and answers the three questions a test asks: what was streamed (`.text`), how it ended (`.done`, `.error`), and with what status.

THE .NET LENS

`MockTransport` is a fake `HttpMessageHandler`: a function from `HttpRequestMessage` to `HttpResponseMessage`, injected under the client. The donor's `ScriptedHandler` is the same thing in thirty lines of C#; here it is a closure.

`monkeypatch.setattr(turns, "request_hash", pinned)` — used once, to simulate a true replay — is the one place a test reaches into a module and replaces a function, which `NSubstitute` would need an interface for.

TRY IT

Play a turn and watch it arrive. Put your OpenRouter key in `.env` as `LUSTJINN_OPENROUTER_API_KEY`, run the migrations and the seed script again, start the server, and create a story with the dummy character on `/docs` as in chapter 2. Then, with the token and the story's id, in a terminal (`-N` keeps curl from buffering):

```
curl -N -X POST http://127.0.0.1:8000/stories/<id>/send \
  -H "Authorization: Bearer <token>" -H "Content-Type: application/json" \
  -d '{"text": "I come in from the rain and shake out my coat."}'
```

The `delta` events appear as the model writes; the `done` event ends them. Open the story on `/docs`: both messages are there. Then:

- Send `/ask` What does she make of me so far? — an answer, and nothing new in the story.
- Send `/aks` anything — refused, with nothing stored and nothing billed.

- Reroll with `POST /stories/<id>/reroll`, then
 `SELECT sequence, role, deleted_at FROM messages` in `psql`: the old reply is hidden, its number is not reused, and `SELECT kind, cost FROM spend` shows every call that was made.
Finally, change the key in `.env` to something wrong, restart, and send again: the `error` event says the message was kept, and `GET /stories/<id>` agrees.

3.9 What step 3 leaves behind

- A streaming OpenRouter client with one error type, one timeout, provider routing, and money read as `Decimal` from the response — never computed from a price list.
- A ledger with a row for every billed call, insert-only, surviving everything.
- A turn that keeps the reader's message before the model is asked, is safe to retry, and streams the reply to the client over server-sent events.
- Reroll, and slash commands where a typo costs nothing. `/ask` answers out of character and stores nothing in the story.
- A scripted model that speaks OpenRouter's stream format, so every test runs without a network.

Next — Chapter 4, The library. Characters with their openings, personas and snippets, with optimistic concurrency and a history of every saved version — and the first foreign keys the owner edits through.

CHAPTER 4

The library

Step 3 played a story with a character that a script had seeded. Step 4 gives the owner the editor: characters with their openings, personas and snippets, on three shelves behind one set of endpoints, with the rules that make editing safe — a save that cannot quietly undo another one, a history that keeps every version, and a delete that the database itself refuses while a story still needs the entry.

This is the chapter on **constraints**: what the schema enforces on its own, what the ORM enforces on every write, and the little that is left for code to check. Along the way, the first mixin, a generic dataclass, and what `UPDATE ... WHERE version = ...` buys.

4.1 Three shelves, one shape

A character, a persona and a snippet are the same shape of thing: a name, a page of text, a version, a time of last change. A character also has an opening. In the donor they were four folders of text files; here they are three tables that share four columns through a **mixin** (`src/lustjinn/models.py`):

```
class LibraryEntry:
    id: Mapped[uuid.UUID] = mapped_column(primary_key=True, default=new_id)
    name: Mapped[str] = mapped_column(String(200))
    version: Mapped[int] = mapped_column(server_default="1")
    updated_at: Mapped[datetime] = mapped_column(
        DateTime(timezone=True), server_default=func.now(), onupdate=func.now()
    )

class Character(LibraryEntry, Base):
    __tablename__ = "characters"
    __table_args__ = (Index("uq_characters_name_lower", text("lower(name)"), unique=True),)

    card: Mapped[str] = mapped_column(Text)
    opening: Mapped[str | None] = mapped_column(Text)

class Snippet(LibraryEntry, Base):
    __tablename__ = "snippets"
    __table_args__ = (Index("uq_snippets_name_lower", text("lower(name)"), unique=True),)

    text: Mapped[str] = mapped_column(Text)
```

`LibraryEntry` is not a table. It is a plain class whose annotated attributes SQLAlchemy copies into each class that inherits it — every table gets its own `id`, `name`, `version` and `updated_at` columns. The unique index on `lower(name)` stays on each class, because it names the table.

THE .NET LENS

A mixin is the abstract base entity that EF Core maps with TPC (table per concrete type): each derived class gets a full table with the base's columns, and the base itself has none. The difference is that nothing here is polymorphic — there is no query over "all entries" — which is why it is a mixin rather than a mapped base. Multiple inheritance, `class Character( LibraryEntry, Base )`, is ordinary Python; the mixin goes first so its attributes win.

The endpoints are written once, for a **shelf**, and the shelf is a value (`src/lustjinn/library.py`):

```
@dataclass(frozen=True)
class Shelf[E: LibraryEntry]:
    slug: Slug
    kind: str
    model: type[E]
    text_of: Callable[[E], str]
    set_text: Callable[[E, str], None]

SHELVES: dict[str, Shelf[Any]] = {
    "characters": Shelf("characters", "character", Character, _card, _set_card),
    "personas": Shelf("personas", "persona", Persona, _text, _set_text),
    "snippets": Shelf("snippets", "snippet", Snippet, _text, _set_text),
}
```

`Shelf[E: LibraryEntry]` is a **generic class** with a bounded type parameter, in the syntax Python 3.12 introduced: `E` is any subclass of `LibraryEntry`, and `model: type[E]` is the class itself, used as `select(shelf.model)` and `shelf.model(name=...)`. The two callables answer the one way the three tables differ — a character's text is its `card`, the others' is `text` — without a conditional at every use.

The path parameter picks the shelf, and its type is a `Literal` of the three slugs:

```
Slug = Literal["characters", "personas", "snippets"]

def shelf_named(shelf: Slug) -> Shelf[Any]:
    return SHELVES[shelf]

OnShelf = Annotated[Shelf[Any], Depends(shelf_named)]

@router.get("/{shelf}/{entry_id}")
async def read_entry(shelf: OnShelf, entry_id: uuid.UUID, session: Session) -> EntryOut:
    ...
```

`GET /library/openings` answers 422 before any code runs: FastAPI validated the path against the `Literal`. And `shelf_named` is a dependency with a parameter of its own, which FastAPI reads from the path — a dependency that takes a path parameter is nothing special.

THE .NET LENS

`Shelf[E]` is `Shelf<E>` where `E : LibraryEntry`, and `type[E]` is `Type` with the generic argument kept — closer to a `Func<E>` factory plus a `typeof(E)` than to anything C# writes in

one token. The `Literal` on the path is a route constraint, `{shelf:regex('^characters|...$')}`, with the benefit that pyright knows the three values too.

4.2 Names

Four rules, each with a reason:

- **Unique without regard to case**, by the index on `lower(name)` from step 2. The endpoint checks first, to answer 409 with a sentence instead of a constraint name, but the index is what holds.
- **Refused:** `< > : " / \ | ? *` and control characters, on every shelf. A name may become a file name when a story is exported (step 7), and being refused a colon is a smaller surprise than an export that only works on one machine.
- **A snippet's name is one word** of letters, digits, `_`, `+` and `-`, up to 32 characters — because it is typed as `:name`, and anything else could not be.
- **A name starting with `_` is kept but not listed** unless asked for. A shelf holds working papers as well as things to play — a template, notes towards a character — and they belong beside what they are about rather than in every picker.

The last rule met a trap worth a box.

WATCH OUT

`select(Snippet).where(~Snippet.name.startswith("_"))` compiles to `name NOT LIKE '_%'`, and in SQL's `LIKE`, `_` matches *any one character*. The condition excluded every name. SQLAlchemy has the fix built in — `startswith("_", autoescape=True)` — but nothing warns when it is left out. A test that created two entries and listed one caught it; a list that happened to be tested with no hidden entries would not have.

The same rule had a second trap, and this one only showed on another machine.

WATCH OUT

The shelf is listed alphabetically. The first version said `ORDER BY lower(name)`, and its test passed where it was written. On the laptop's Postgres the same test failed: `_template` came out *between* Elena and Zoe.

`ORDER BY` on text does not have one meaning. Postgres sorts by the **collation** the database was created with, and collations disagree: the usual `en_US.utf8` ignores punctuation on a first pass, so a leading underscore vanishes; the `C` collation compares bytes, so `Ángela` lands after `Zoe`. The fix names the order instead of inheriting it:

```
def alphabetical(name: InstrumentedAttribute[str]) -> ColumnElement[str]:
    return name.collate("und-x-icu")
```

`und-x-icu` is ICU's language-neutral order — punctuation, digits, then letters without regard to case, accents beside their base letter — and it is the same on every server. A test now pins `_template`, `9lives`, `alba`, `Ángela`, `Elena`, `Zoe` exactly.

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SQL Server has the same hazard under another name: a column's collation (SQL_Latin1_General_CP1_CI_AS and friends) decides both ordering and equality, and a query that sorts one way on your machine sorts another on a server installed with a different default. `.collate(...)` is `EF.Functions.Collate(x, "...")`.

4.3 What a foreign key does for you

A story points at its character and its persona by id, `ON DELETE RESTRICT`. Two things follow, and neither needs code:

- **Renaming is free.** The donor stored the character's *name* in each conversation, so a rename broke every story using it. Here a `PATCH` with a new name reaches every story at once, and a new card reaches them from their next turn —
`test_renaming_a_character_a_story_uses_reaches_the_story` sends a turn after the rename and reads the new card in the prompt.
- **Deleting one a story uses is refused**, by Postgres, whoever asks.

The code's part is to say *which* stories, so the refusal is useful: `used_by` lists the live stories using an entry, and `DELETE` answers 409 with their names. It is asked at the moment of deleting, not trusted from when a screen was drawn — a story could have been started with it in between.

One case the key cannot express: a **deleted** story (`deleted_at` set) no longer counts as a user, but its row still points at the character, and the foreign key does not know the difference. So `used_by` says nobody, the delete goes to the database, the database refuses, and the endpoint turns that `IntegrityError` into its own 409: *a deleted story still refers to this character; it can go once that story is erased for good* — which is step 7's purge.

```
await session.delete(entry)
try:
    await session.commit()
except IntegrityError:
    await session.rollback()
    raise HTTPException(409, "A deleted story still refers to this ...") from None
```

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`IntegrityError` is `DbUpdateException` wrapping a `PostgresException` with SQLSTATE 23503. The pattern is the same: let the database be the last word, and translate its refusal where the caller can act on it. `raise ... from None` drops the inner exception from the traceback — the equivalent of throwing a new exception without an `InnerException`, on purpose, because the database's message is not the reader's business.

4.3.1 The default persona

A story that names no persona plays as the **default**, which the owner sets once. It lives in a new table, `settings`, with one row:

```
class AppSettings(Base):
    __tablename__ = "settings"
    __table_args__ = (CheckConstraint("id = 1", name="one_row"),)
```

```

id: Mapped[int] = mapped_column(primary_key=True, default=1)
default_persona_id: Mapped[uuid.UUID | None] = mapped_column(
    ForeignKey("personas.id", ondelete="SET NULL")
)

```

A `CHECK (id = 1)` is how a table is made to hold one row at most. The foreign key is `SET NULL` rather than `RESTRICT`: deleting the default persona is allowed, and clears the setting — but only when nobody plays as it. The default counts as **used by every story that names none**, and `used_by` says so, so the delete is refused with those stories' names while any exist. That rule is the one piece of "who uses this" that the schema cannot carry, because the link runs through a setting rather than a column.

With no default set, a story that names no persona plays as nobody — even when exactly one persona is on the shelf. Choosing it would be choosing who the reader is playing, and it would change the day a second one appeared.

4.4 Optimistic concurrency

The owner edits a card on the laptop, then an hour later on the phone, from a page opened before the laptop's save. Saving the phone's text would silently undo the laptop's. The donor caught this by hashing the file; here the row has a **version**, and a save names the version it started from:

```

class EntryChanges(BaseModel):
    version: int
    name: Name | None = None
    text: Body | None = None
    opening: str | None = None

```

```

entry = await _one(session, shelf, entry_id)
if changes.version != entry.version:
    raise _changed_elsewhere(shelf, entry)

```

The refusal is a 409 that carries the entry **as it is now**, so a client can show both texts and let the reader decide — nothing is lost either way, which is the whole point. Saving again with the current version succeeds, and the overwritten text is in the history.

That check compares the client's version with the row as loaded. Between that load and the `UPDATE`, another save could still land. The ORM closes that gap by itself, with one mapper option on the mixin:

```

@declared_attr.directive
@classmethod
def __mapper_args__(cls) -> dict[str, Any]:
    return {"version_id_col": cls.version}

```

With `version_id_col` set, every `UPDATE` the ORM writes for these tables carries `WHERE version = <the version it loaded>` and sets `version = version + 1`. If no row matched, someone saved first, and the flush raises `StaleDataError` — which the endpoint answers with the same 409. `test_the_orm_checks_the_version_on_every_update_it_writes` bumps the version behind the session's back with raw SQL and watches the flush fail.

THE .NET LENS

`version_id_col` is `[ConcurrencyCheck]` on the property, or `IsConcurrencyToken()` in `OnModelCreating`, and `StaleDataError` is `DbUpdateConcurrencyException`. One difference: EF compares the *original* value it tracked; SQLAlchemy increments the column itself and compares the number. Either way the write carries the check, so there is no window.

`@declared_attr.directive` marks a class-level setting a mixin computes per subclass — needed here because `cls.version` is a different column on each table. In EF the same thing would be a line in a base `OnModelCreating` helper called for each entity.

A save that changes nothing writes nothing: `session.is_modified(entry)` is false, no `UPDATE` is issued, the version stays, and no history row is added. The donor's test for this — `Saving_the_text_unchanged_does_not_replace_the_backup` — has its namesake in `tests/test_library_history.py`.

4.5 History

The donor kept one `.bak` of the last save. Here every version is kept, in `library_history`: kind, the entry's id, the version number, the name, text and opening as saved, and when. A row is written on create (version 1) and on every save that changed something, in the same transaction as the change:

```
if session.is_modified(entry):
    session.add(_remembered(shelf, entry, version=entry.version + 1))
    await session.commit()
```

Two decisions carry over from the ledger. **No foreign key**: the history of a deleted entry is still history, and `GET /library/personas/{id}/history` answers for an id that is no longer on the shelf. **Insert-only**, by a trigger (migration 0006) that refuses `UPDATE`, `DELETE` and `TRUNCATE`. The history is the one thing in the library that cannot be edited.

WORTH KNOWING

Three tables now share the insert-only trigger pattern: `messages` (with a purge door), `spend` and `library_history` (without). It is the same twelve lines of PL/pgSQL each time, and that is fine — a trigger is schema, and schema is written out. The course mentions it so the repetition does not look like an oversight.

4.6 Snippets expand at send time

A snippet is authored prose deployed at the dramatically right moment: written once, typed as `:storm`, replaced by its text before the message is stored. `src/lustjinn/snippets.py` does the replacing, and the hard part is not finding a colon but knowing when a colon is *not* a trigger. Prose is full of them — `10:30`, `https://...`, `note: this` — and a message is permanent, so guessing is not allowed. The rules:

- The colon must **open a word**: preceded by whitespace or nothing. That excludes clock times and URLs.

- The name is letters, digits, `_`, `+`, `-`, up to 32 of them. A longer run is not a name cut short; it is not a name.
- A closed `:name:` is an **emoji shortcode**, which the clients handle, and the API passes through untouched. This was the open question on the issue; the decision is that emoji are presentation, not content, and belong where the typing happens.
- An unknown name is left exactly as typed. One pass: an expansion is not scanned again.

The expansion runs first in `send`, before the line is read for commands — so a trigger works inside a question too — and before the request hash, so a retry of `:storm` is still one turn. The stored text is the expanded text; a later edit of the snippet does not reach it, which is what “copied, never read again” means.

```
def expand(text: str, snippet: Callable[[str], str | None]) -> str:
    out: list[str] = []
    i = 0
    while i < len(text):
        opens_word = text[i] == ":" and (i == 0 or text[i - 1].isspace())
        if not opens_word:
            out.append(text[i])
            i += 1
            continue
        end = i + 1
        while end < len(text) and text[end] in NAME_CHARACTERS and end - i <= MAX_NAME_LENGTH:
            end += 1
        name = text[i + 1 : end]
        ...
```

`expand` takes the lookup as a function, `Callable[[str], str | None]`, rather than a table. The endpoint hands it a closure over the snippets it loaded; the tests hand it a dictionary through `by_name`. A function parameter is the cheapest seam there is.

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`Callable[[str], str | None]` is `Func<string, string?>`, and the scanner is a line-for-line port of the donor’s `ShortcodeScanner.ExpandAll` with the emoji branch removed. The one Python idiom worth noticing is `text[i + 1 : end]` — slicing, half-open like `Range` in C#, and the spaces around the colon are what ruff’s formatter does when the bounds are expressions.

4.7 What the tests look like now

The library tests are the first to use the API’s own endpoints as the *arrange* step — a character created through `POST /library/characters` rather than inserted through the session — because the endpoints are now the natural way to make one. The two styles coexist: `a_character(session)` where the test is about something else, the endpoint where the test is about the library.

One test needed a line of care:

```
character_id = character.id # the refused delete rolls the shared session back
```

The test and the app share one session, and the app’s `rollback()` after the refused delete expires every object in it — including the test’s own `character`. Reading `character.id` afterwards would try

to reload it, which async code cannot do on its own. Capture what you need before the request; a test that shares a session with the code it tests shares its side effects too.

TRY IT

Rebuild the database and seed it, then edit the dummy from `/docs`:

1. GET `/library/characters` — one entry, "The Gilded Heron", version: 1.
2. PATCH it with `{"version": 1, "name": "The Heron"}`. The response says version: 2.
3. PATCH it again with `{"version": 1, "text": "x"}` — 409, and `current` is what it says now. Send `{"version": 2, "text": "x"}` instead: 200, version: 3.
4. GET `.../history` — three rows, newest first, with the card you just overwrote.
5. DELETE it: 409, naming the stories that use it. Delete the story first, then try again: 409 again, from the foreign key — a deleted story still points at it. That is the purge's job, in step 7.
6. POST `/library/snippets` with `{"name": "storm", "text": "Rain hammers the glass."}`, then send a turn whose text is `I come in. :storm` — the stored message carries the expansion. Send at `10:30 :nothing` — stored as typed.

4.8 What step 4 leaves behind

- Three shelves behind one set of endpoints, with names unique without regard to case, hidden working papers, and the default persona that a story without one plays as.
- Renames that reach every story at once, and deletes the database refuses while a story needs the entry — with the stories named.
- Saves that carry the version they started from, checked twice: by the endpoint against the client's version, and by the ORM on the write itself.
- A history of every version, insert-only, that outlives the entry.
- Snippets that expand where `:name` opens a word, and nowhere else.

Next — Chapter 5, The context builder. The prompt, properly: layers in order from least to most volatile, a token budget counted with the model's own vocabulary, the newest turn always kept, and an audit of what each layer cost — mostly pure functions, and mostly tests.

CHAPTER 5

The context builder

Until now the prompt was three things in a row: the card, the persona, the whole transcript. That works for a story of twenty turns. It does not work for a story of two hundred, and the thesis of this project — *you do not need an infinite context window if you have a good retrieval layer* — needs a place where the prompt is *assembled* rather than concatenated. Step 5 builds that place: a module of pure functions that takes everything a prompt could hold and returns what the model is sent, within a budget, with an account of what it did.

Nothing in this step touches the database or the network. That is the point of it, and the lesson: the most important logic in the app is the easiest to test.

5.1 Why layers, and why this order

A model is sent a list of messages. The order of that list is a cache key: providers keep a **prefix cache**, and a request whose first n messages are byte-for-byte the same as the last request's gets those n messages for a fraction of the price. So the prompt is laid out from the thing that never changes to the thing that changes every turn:

Layer	Changes	Here since
character	when the card is edited	step 3
persona	when the persona is edited	step 3
directives	when a dial is turned	step 7
world	when a fact is extracted or retired	step 6
summaries	every few dozen turns	step 6
history	every turn — and it is the layer that <i>gives way</i>	step 3
memories	every turn	step 6
trackers	every turn	step 7
instruction	every turn; this turn only	step 3

Retrieval goes *after* the history on purpose. Recalled turns change with every message, and putting them before the transcript would invalidate the cache from that point down, every turn.

5.2 The shape: dataclasses in, dataclasses out

`src/lustjinn/context.py` has one input type and one output type:

```
@dataclass(frozen=True, kw_only=True)
class Layers:
    character: Character
    persona: Persona | None = None
    directives: str | None = None
    facts: Sequence[Fact] = ()
    summaries: Sequence[str] = ()
    history: Sequence[Message] = ()
    memories: Sequence[Recalled] = ()
    trackers: str | None = None
    instruction: str | None = None

@dataclass(frozen=True)
class Built:
    messages: list[ChatMessage]
    spent: dict[str, Spent]
    budget: int | None
```

A **dataclass** is a class whose `__init__`, `__repr__` and `__eq__` are written for you from the annotated fields. `frozen=True` makes instances immutable. `kw_only=True` means every field must be passed by name — `Layers(character=c, history=h)`, never `Layers(c, h)`.

The keyword-only rule is not taste. `airp`'s `Build` took its layers positionally, and three of its bugs came from the same move: a new layer added in the middle of the parameter list, and one call site that kept passing the old arguments in the old positions — summaries where the directives should be, compiling cleanly. Names cannot be shifted.

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`@dataclass(frozen=True)` is a record with `init`-only properties; `kw_only=True` is a constructor you may only call with named arguments. `Sequence[Message]` is `IReadOnlyList<Message>` — a type that promises you will only read. `Fact` and `Recalled` are small records that steps 6 and 7 will produce; they exist now so the builder's shape does not change when they arrive.

5.3 Counting tokens

A budget is measured in tokens, and there is exactly one honest way to count them: the way the model does. `src/lustjinn/tokens.py`:

```
ENCODING = "o200k_base"
PER_MESSAGE = 4

@lru_cache
def encoding() -> tiktoken.Encoding:
    return tiktoken.get_encoding(ENCODING)

def count(text: str) -> int:
```

```

return len(encoding().encode(text, disallowed_special=()))

def for_message(text: str) -> int:
    return count(text) + PER_MESSAGE

```

tiktoken is OpenAI's tokenizer library; **o200k_base** is the vocabulary of their current models, and the one airp counted with. Two details:

- **PER_MESSAGE = 4**: a message costs more than its text. The role and the separators the API wraps around it come to about four tokens — a figure measured against what a provider reported for a real transcript, not derived from anything.
- **disallowed_special=()**: by default tiktoken refuses to encode text containing a marker like `<|endoftext|>`, because in a prompt those mean something. In a card they are just characters. A character sheet that quoted one would have crashed the count.

WATCH OUT

A characters-per-token constant is not a count. Spanish runs about 30% more tokens per character than English in these vocabularies; a budget that holds in one language overflows in the other. That is why there is a tokenizer in the project and not a division by four.

WORTH KNOWING

tiktoken downloads the vocabulary file (3.6 MB) the first time it is used and keeps it in a cache directory — `TIKTOKEN_CACHE_DIR` if set, the system's temporary folder otherwise. On this machine and in CI that happens once and silently. A sandbox that blocks `openaipublic.blob.core.windows.net` cannot count tokens until the domain is allowed.

5.4 The budget

`build(layers, budget=32_000, recall_percent=10)` does its arithmetic in a fixed order:

1. **Fit the memories under their cap.** The cap is $\text{budget} \times \text{recall_percent} / 100$ — a *share*, not a count. Four recalled turns once filled a 60,000-token prompt, which is why the rule is not “at most four”. Memories arrive most relevant first; each is kept if the layer still fits under the cap with it, and one that does not fit is **skipped, not fatal** — a shorter one further down may still make it. The survivors are then sorted back into story order, so the model reads them as a transcript.
2. **Cost every fixed layer.** Character, persona, directives, world, summaries, memories, trackers, instruction: these go in whole, whatever they cost. The instruction is counted here too — it is never squeezed out by old turns.
3. **Give the history what is left.** `room = max(0, budget - fixed)`. Walk the transcript from the newest message backwards, adding each turn while it fits, and **stop at the first that does not**. The kept turns are always a contiguous run ending on the newest.
4. **The newest message is always kept**, before the room is even checked. A prompt that leaves out the words the reader just typed is a reply to nothing — and a real story once lost the reader's own message that way. If the budget cannot hold even the fixed layers, the prompt goes *over budget* by one message rather than under it by the one that matters.

```
def _fit_history(history: Sequence[Message], room: int | None) -> tuple[list[Message], int]:
    if not history:
        return [], 0
    kept = [history[-1]]
    used = tokens.for_message(history[-1].text)
    for turn in reversed(history[:-1]):
        cost = tokens.for_message(turn.text)
        if room is not None and used + cost > room:
            break
        kept.append(turn)
        used += cost
    kept.reverse()
    return kept, len(history) - len(kept)
```

The default budget is 32,000 tokens — far below any current model’s window, on purpose. Two reasons, both measured on real stories: attention thins out as a prompt grows, so a longer transcript does not make a better reply past a point; and every token in the prompt is paid for on every turn, so a 100,000-token prompt costs three times what a 32,000-token one does, forever.

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`_fit_history` is the loop you would write with a `for` over `history.Reverse().Skip(1)` and a running total. Python’s `reversed(history[:-1])` is the same `Reverse().Skip(1)` read inside-out: slice off the last element, then iterate backwards. The tuple return `(kept, dropped)` is a `(List<Message> Kept, int Dropped)` value tuple; the caller unpacks it as `history, dropped = ...`.

5.5 The audit

Every `Built` carries `spent`: per layer, how many tokens it took and how many items were dropped to fit. `describe()` turns that into one line:

```
character 2755 · persona 412 · history 1830 (12 dropped) · instruction 60 · total 5057/32000
```

That line is stored on every reply, in `messages.context_audit`, with the total in `estimated_prompt_tokens` — right beside `prompt_tokens`, the figure the provider actually billed. `GET /stories/{id}/audit` lists the newest twelve replies and eight questions with both numbers. Rerolled replies are listed too, marked hidden: “why did it say that?” is asked of them more than of any other.

Having the estimate beside the bill is what made the next paragraph possible.

WATCH OUT

The first real turn of this project was billed at **2,755** prompt tokens. The counter said **3,163** for the same prompt — 15% high. Not a bug in the arithmetic: `o200k_base` is OpenAI’s vocabulary, and DeepSeek has its own, which happens to spend fewer tokens on this prose. High is the safe direction (a prompt the counter thinks is at the budget is under it), so the test that checks the real turn pins the margin at “between the billed figure and 20% above it” rather than pretending to 10%. Closing the gap is #94, and the audit is collecting the data for it on every reply.

5.6 Testing pure logic

Thirty-five tests cover this step, and most run in under a millisecond, because the thing under test is a function from values to values. The tests build their inputs by hand:

```
def turns(*lines: str) -> list[Message]:
    roles = [Role.USER, Role.ASSISTANT]
    return [Message(sequence=i + 1, role=roles[i % 2], text=line) for i, line in enumerate(lines)]

def test_the_turns_kept_are_the_ones_nearest_the_reply(dummy: Dummy) -> None:
    built = build(
        Layers(character=heron(dummy), history=long_story(100)),
        budget=tokens.for_message(dummy.card) + 500,
    )

    kept = [m.content for m in built.messages[1:]]
    assert kept[-1].startswith("Turn 100.")
    numbers = [int(line.split(".")[0].split()[1]) for line in kept]
    assert numbers == list(range(numbers[0], 101))
```

Three things to notice:

- **Message(...)** without a database. A SQLAlchemy model is still a Python object; constructing one costs nothing and saves nothing. The test never opens a session.
- **The card is the dummy's**, 1,850 words, through the `dummy` fixture. `airp`'s equivalent tests used a four-token card inline, and a bug that only appeared with a real-size card cost a story twenty-four turns of memory. Here the fixed layers are big enough to make the budget bind the way it does in a real story.
- ***lines: str** is `params string[] lines: turns("One.", "Two.", "Three.")`.

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These are `xUnit [Fact]`s over a static class, and the fixture is `IClassFixture<Dummy>`. What has no counterpart is how little ceremony it takes: no `Arrange` helpers to inject, no `IOptions<ContextOptions>` to fake. `build` takes its budget as an argument, so a test passes a small number and the rule shows itself.

TRY IT

Play a turn, then look at what it was built from:

```
GET /stories/{id}/audit
```

Compare `estimated_prompt_tokens` with `prompt_tokens` on the newest reply. Then set `LUSTJINN_CONTEXT_BUDGET=4000` in `.env`, restart the server, play another turn, and look again: `history` now shows `(n dropped)`, and the total sits just under 4,000 — unless the card alone is bigger than that, in which case the total is over and the history is exactly one message. Put the budget back.

From `psql`, the same comparison across every reply so far:

```
SELECT sequence, estimated_prompt_tokens, prompt_tokens,  
       round(100.0 * estimated_prompt_tokens / prompt_tokens) AS estimate_pct  
FROM messages WHERE prompt_tokens IS NOT NULL ORDER BY sequence;
```

5.7 What step 5 leaves behind

- One place where the prompt is assembled, in a fixed order, with slots for everything steps 6 and 7 will add.
- A token count in the model's own units, and a budget the transcript yields to — oldest first, newest never.
- Recalled memories capped at a share of the budget, ready for retrieval to fill.
- An audit on every reply, estimate beside bill, and the first calibration it produced.

Next — Chapter 6, The memory. The three mechanisms that let a story outlive its budget: summaries of what happened, retrieval of what was said, facts about what is true now — and the background calls that produce them only when a story needs them.